

CORONARY CIRCULATION

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- **Coronary Arteries :—**

- Heart muscle is supplied by **2 coronary arteries** (*Right Coronary Artery and Left Coronary Artery*).
- Arteries encircle the heart in the manner of a crown, hence named **coronary arteries**.

ARTERIAL SUPPLY & DOMINANCE

- a. **Right Coronary Artery (RCA)** : Supplies the whole of the right ventricle and the posterior portion of the left ventricle.
- b. **Left Coronary Artery (LCA)** : Supplies mainly the anterior and lateral parts of the left ventricle.

Coronary Distribution Pattern	Prevalence in Humans
Right Coronary Artery Large (Right Dominant)	50 – 60% of humans
Left Coronary Artery Large (Left Dominant)	15 – 20% of humans
Both Arteries Almost Equal (Balanced / Co-dominant)	20 – 30% of humans

- **Branches of Coronary Arteries :—**

Coronary Arteries → Smaller branches: **Epicardiac Arteries** → Final branches: **Intramural Vessels**

VENOUS DRAINAGE

- Drainage occurs via **3 types of vessels**:
 - a. **Coronary Sinus :—** Large vein draining **75% of total coronary flow**. Drains blood from the left side and opens into the **right atrium** near the tricuspid valve.
 - b. **Anterior Coronary Veins :—** Drain blood from the right side and **open directly into the right atrium**.
 - c. **Thebesian Veins (Venae Cordis Minimae) :—** Drain deoxygenated blood from the myocardium **directly into the concerned chamber of the heart**.

- **Physiological Shunt :—**

The diverted route (diversion) through which venous (deoxygenated) blood is mixed directly with arterial blood (*e.g., drainage of deoxygenated blood from bronchial circulation and thebesian veins into pulmonary vein / left side of heart*).

FACTORS REGULATING CORONARY BLOOD FLOW

- **Autoregulation** : Capacity of the heart to regulate its own blood flow within a wide range of perfusion pressure.
- **Key Regulatory Factors** :—
 - a. Need for oxygen
 - b. Metabolic factors
 - c. Coronary perfusion pressure
 - d. Nervous factors

1. NEED FOR OXYGEN

- Amount of blood passing through the coronary circulation is **directly proportional to the consumption of oxygen** by cardiac muscle.
- **Mechanism** : Hypoxia immediately causes **coronary vasodilation** → ↑ increases blood flow.

2. METABOLIC FACTORS

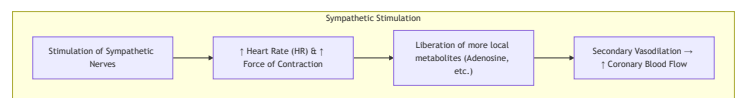
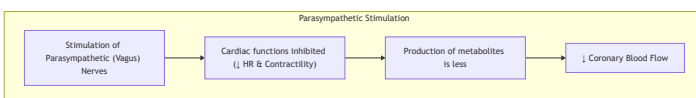
- **Reactive Hyperemia** : ↑ Increase in coronary blood flow due to vasodilator effects of local metabolites.
- **Metabolic Products Causing Coronary Vasodilation** :—
 - a. **Adenosine** (*Most potent vasodilator*)
 - b. **Potassium ions** (K^+)
 - c. **Hydrogen ions** (H^+)
 - d. **Carbon dioxide** (CO_2)
 - e. **Adenosine phosphate compounds** (AMP, ADP)

3. CORONARY PERFUSION PRESSURE

- Determined by the balance between **Mean Arterial Pressure (MAP)** and **Venous Pressure**.

4. NERVOUS FACTORS

- Coronary blood vessels are innervated by both **parasympathetic and sympathetic divisions** of the Autonomic Nervous System (ANS).



APPLIED PHYSIOLOGY

Condition	Definition & Pathophysiology
1. Coronary Occlusion	Partial or complete obstruction of a coronary artery.
2. Myocardial Ischemia	Reaction of a part of the myocardium in response to hypoxia (inadequate blood supply).
3. Myocardial Infarction (Heart Attack)	Necrosis of the myocardium caused by insufficient blood flow due to an embolus, thrombus, or vascular spasm .
4. Cardiac Pain (Angina Pectoris)	• Chest pain caused by myocardial ischemia. • Common clinical manifestation of coronary artery disease.